

Solutions – Mid Semester Test (SE ZG622: Software Project Management)

Q1. Horizontal vs. Vertical Story Splitting in Scrum (6 Marks)

User Story: “As a customer, I want to transfer funds from my account to another account.”

Horizontal Story Splitting (Layer-wise):

- Split by technical layers/components (UI, backend, database).

Examples:

1. Create UI screen for entering transfer details (account number, amount).
2. Implement database schema for transaction logging.
3. Develop backend API for transfer validation.

Limitation: Deliverables are not end-to-end usable by the customer.

Vertical Story Splitting (Feature-wise):

- Split by functionality that provides customer value end-to-end.

Examples:

1. Transfer between my own accounts (within the same bank).
2. Transfer to another account within the same bank.
3. Transfer to an account in another bank (via NEFT/RTGS/IMPS).

Benefit: Each slice delivers a working feature that the customer can use.

■ **Conclusion:** Vertical slicing is preferred in Scrum as it delivers incremental customer value, while horizontal slicing focuses on internal development layers.

Q2. Direct Measures in Software Measurement (6 Marks)

Direct Measures (measured without models/derivations):

1. Lines of Code (LOC):

- Contribution: Helps track project size, productivity (LOC/day), and maintenance effort.
- Collection: Automated tools like SonarQube, IDE plug-ins can count LOC.

2. Execution Time (Performance Measure):

- Contribution: Indicates system responsiveness, critical for performance-sensitive projects.
- Collection: Profiling tools, performance testing frameworks (e.g., JMeter).

■ **Role in Project Management:** Direct measures provide quantitative data for tracking progress, productivity, and quality, assisting managers in project planning and corrective actions.

Q3. Role of Project Manager: Plan-Driven vs. Agile (6 Marks)

Plan-Driven (e.g., Waterfall):

- Creates detailed upfront project plan (scope, schedule, cost).
- Monitors adherence to plan, manages risks, controls deviations.
- Heavy focus on documentation, reporting, and approvals.
- PM = Controller of the plan.

Agile (e.g., Scrum):

- Facilitator and servant leader for the team.
- Focuses on enabling collaboration, removing impediments, and ensuring value delivery.
- Planning is adaptive: sprint planning, daily stand-ups, backlog grooming.
- Shared responsibility with Product Owner and Scrum Master.
- PM = Enabler and coach.

■ **Key Difference:** Plan-driven PM is directive and control-oriented, Agile PM is facilitative and adaptive.

Q4. Story Points vs. Function Points (6 Marks)

Story Points (Agile):

- Definition: Relative measure of effort, complexity, and risk.

Example:

- Transfer between my accounts = 2 SP
- Transfer to other bank = 8 SP
- Usage: Helps in sprint planning and velocity tracking. No fixed unit.

Function Points (Plan-driven):

- Definition: Standardized measure of software functionality delivered (independent of technology).

Example:

- Input screen for account details = 3 FP
- Output confirmation = 2 FP
- Interface to external bank = 5 FP
- Usage: Used for cost estimation, productivity metrics, contractual commitments.

■ Difference: Story points are team-specific and subjective; function points are objective, standardized, and used for formal estimation.

Q5. Measures for External Quality Attributes in IRCTC E-ticketing System (6 Marks)

1. Reliability:

- Measure: Mean Time Between Failures (MTBF), Error Rate in transactions.
- Justification: Ensures users can book tickets consistently without failures.

2. Performance:

- Measure: Average Response Time, Transactions per second (TPS).
- Justification: Crucial during high load (e.g., Tatkal booking).

3. Usability:

- Measure: Task completion rate, User satisfaction survey score, Clicks to complete booking.
- Justification: Determines how easily customers can complete ticket booking.

■ These measures allow managers to monitor system quality, prioritize improvements, and align with user expectations.